

Economic History II

Caravan trade and lost cities

FGV EPGE

Question

Around 1880 BC merchants from Assur ran a caravan trade into Anatolia, and wrote almost all of it down. What can be learned from their paperwork?

- About 23,000 clay tablets, dug out of private houses at Kanesh.
- Letters, contracts, debt notes, expense accounts: the working papers of a business.
- Larsen reconstructs the trade from them.
- Barjamovic, Chaney, Coşar and Hortaçsu count the same tablets and put lost cities back on the map.

Today's reading: Larsen, *Ancient Kanesh*

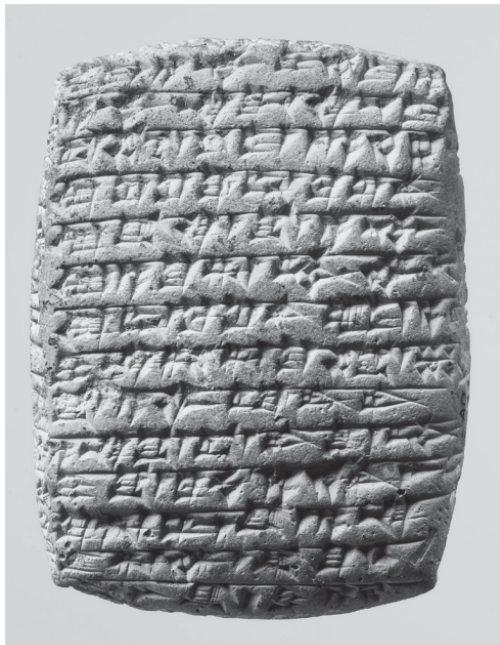
- Mogens Trolle Larsen, *Ancient Kanesh: A Merchant Colony in Bronze Age Anatolia* (2015). Four chapters.
- **Ch. 1**: the letter from Assur-idi to his son Assur-nada, and what kind of evidence this is.
- **Ch. 5**: how the tablets came to be readable, and why the merchants wrote so much themselves.
- **Ch. 14**: the caravan, from the gate of Assur to the palace at Kanesh.
- **Ch. 15**: how much moved, and where tin, textiles and copper came from.

Today's paper

- Gojko Barjamovic, Thomas Chaney, Kerem Coşar and Ali Hortaçsu, *Trade, Merchants, and the Lost Cities of the Bronze Age*, *Quarterly Journal of Economics*, 2019.
- Same tablets Larsen reads, counted instead of narrated.
- Each recorded journey between two cities becomes one observation.
- From those counts: how fast trade fell with distance, where the lost cities were, and how big each city was.

Outline

1. The tablets: what was found at Kanesh, and who wrote it.
2. The caravan trade, in the merchants' own numbers.
3. The paper: from letters to trade data.
4. Lost cities back on the map, and city size, then and now.



One letter

- Assur-idi, an old merchant in Assur, writing to his son Assur-nada in Kanesh.
- *"After you came here, you broke your promises five or six times!"*
- Dried in the sun, wrapped in cloth, sealed in a clay envelope, handed to a donkey driver at the city gate.
- A few weeks on the road, about 1,000 kilometers, delivered in Anatolia.
- Dug up around 1900 AD, sold on the antiquities market, now in the Louvre.

What kind of evidence this is

- Everything comes out of excavations: no manuscript was copied and recopied down to us.
- So the corpus is *“a random sample of mostly everyday practical texts of no literary, historical or philosophical importance”*.
- Nearly ten thousand letters, turning on money, debt and profit.
- The same people recur, so families can be traced over five or six generations.
- No archive survives from Assur itself: the mother city is reconstructed from a colony hundreds of kilometers away.



The setting

- Assur: a small city-state on a cliff over the Tigris, in the north of modern Iraq.
- Kanesh (modern Kültepe): central Anatolia, near Kayseri in Turkey.
- Between them roughly 1,000 kilometers of steppe and mountain.
- Assyrian merchant colonies (*karum*) in some twenty Anatolian towns, plus smaller stations. Kanesh was the central one.
- Out of Assur: tin and textiles. Back from Anatolia: silver.

Understanding the texts

- At first nobody could read them: Pinches in 1881 judged the language “*evidently not Assyrian*”.
- Contenau in 1919, then Julius Lewy, established that it was Old Assyrian, around 2000 BC.
- A thousand years separate these texts from the Neo-Assyrian ones scholars already knew.
- Larsen’s yardstick: Shakespeare is hard enough at four hundred years, and this gap is more like *Beowulf*.

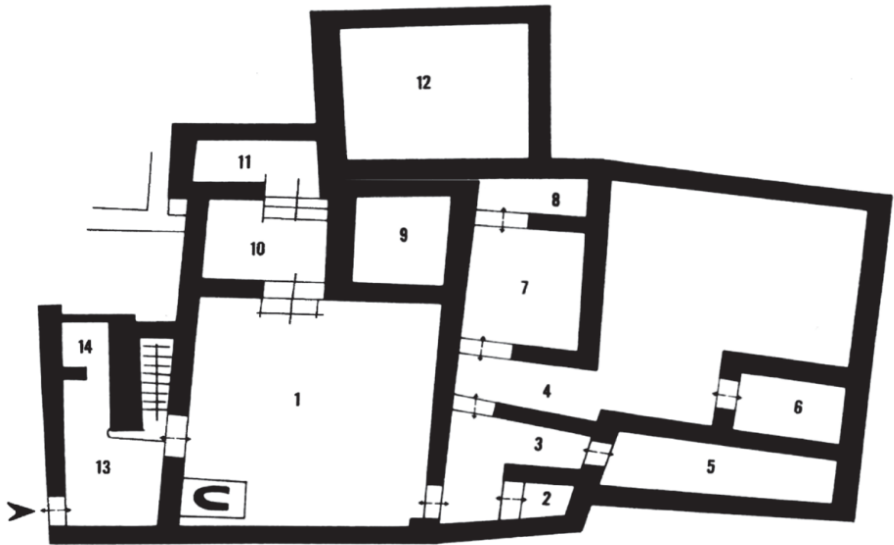
Merchants who wrote their own letters

- In ancient societies usually less than 1 percent of the population was literate.
- Among the Assyrians in Anatolia, most adult men and some women could read and write.
- The script was cut down for them: fewer signs, simpler signs, and a word divider added.
- Larsen: they “**democratised writing**”, making it available to all reasonably intelligent people.
- A merchant far from any scribe had to read a confidential letter himself, and write the reply.

Why were the Assyrians in Anatolia?

- Lewy: conquerors and settlers, an early Assyrian empire reaching into Anatolia.
- Landsberger: merchants operating under the protection of local rulers.
- The argument turned on one word, *annukum*: lead or tin.
- Lewy needed lead, because tin makes bronze and bronze makes weapons.
- Landsberger showed in 1965 that it was tin, and with it the commercial reading of the whole system.

Larsen: *"The Assyrians were in Anatolia as merchants, not as conquerors"*.



12. The house belonging to the Anatolian royal official Peruwa. On the large plan of the lower town it is house 102. The irregular shape and the incorporation of rooms from a level 3 house as cellars (9 and 12) indicates that the building had been enlarged and changed. Kültepe Excavation Archives.

Reading the town house by house

- Up on the mound, the palace and a customs house: burned hot enough to vitrify the bricks, and almost no texts. The record is all from the lower town.
- Archives sat in strongrooms on wooden shelves, and fell to the floor when the houses burned.
- We know who lived where: Usur-sha-Ishtar (1,600 tablets), Elamma (about 800), Shalim-Assur (house 90, about 1,200).
- Assyrians and Anatolians lived side by side, in the same kind of house, with the same pottery.
- Only the archives tell them apart. Ancestors were buried under the floors.

Assur, the mother city

- A small city-state on the Tigris, with a god who is the city itself.
- Royal inscriptions record freedom from taxation on “*silver, gold, copper, tin, barley, wool*”.
- That policy made Assur a transit center: goods arrived from the south and were re-exported to Anatolia.
- Assur produced very little of what it sold. Its business was to sit on a route.

Trade between Assur and Kanesh

Morning at the gate of Assur

- Loaded donkeys checked a last time by their packers; harnesses and saddlebags fastened.
- Officials from the City Hall inspect the packing and the seals, and note whether the export duty is paid.
- Families seeing off sons who will be away for months or years.
- A young woman leaving for Kanesh to marry a man she has probably never met.
- Sealed letters handed to whichever traveler looks reliable.

Weights, not coins

- No coined money here. Coins are invented some 1,200 years later.
- Silver is weighed out for every deal, so the weights are the units.
- **Shekel**: about 8 grams. **Pound** (a mina): 60 shekels, about half a kilo. **Talent**: 60 pounds, about 30 kilos.
- So a donkey carries about 160 pounds, near 80 kilos, and 30 pounds of silver is 15 kilos of it.
- Prices come as ratios by weight: 13 shekels of tin for 1 shekel of silver.

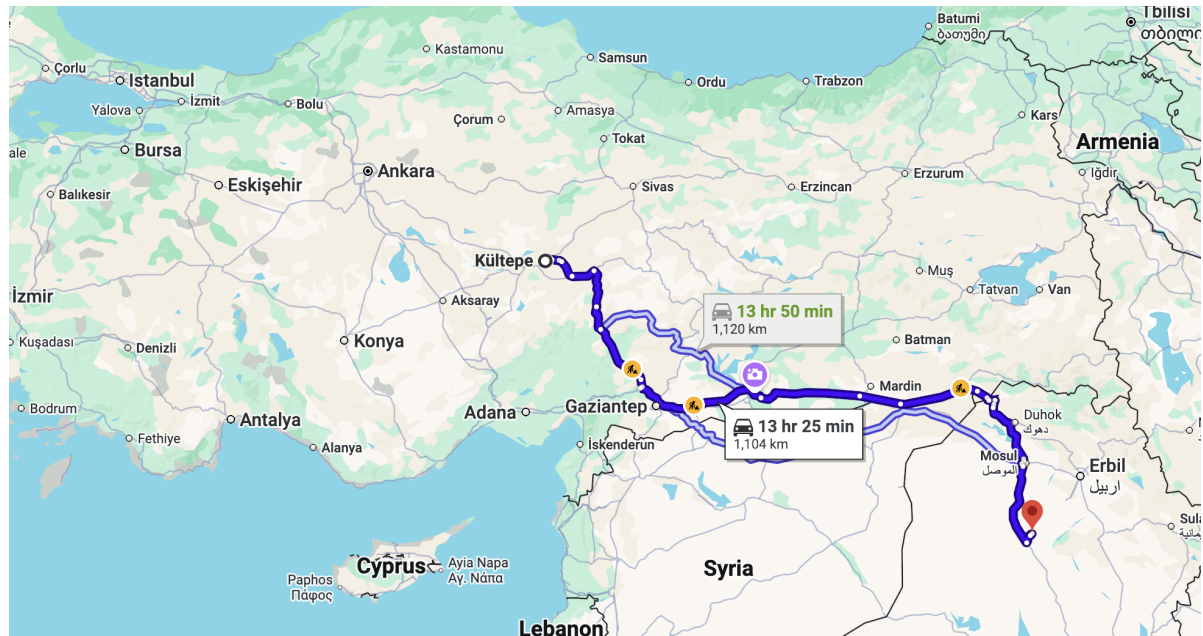
Larsen, ch. 14, p. 187: *"a shekel weighed some 8 grams"*.

What a caravan was

- About twenty donkeys and about as many men.
- Each donkey: two sealed saddlebags plus a top-pack, roughly 160 pounds.
- Tin in bags of 65 pounds; textile bags hold 25 to 35 pieces.
- The cargo belongs to several merchants at once, a few donkeys each.
- The largest one recorded: 34 donkeys, 603 *kutano* textiles, 600 kilos of tin, plus oil, gemstones and saffron.

The road

- About 750 kilometers as the crow flies, 1,000 to 1,100 on the ground.
- Larsen's comparison: Paris to Barcelona, or Washington to Chicago.
- Roughly 30 kilometers a day, so about six weeks.
- The Euphrates by ferry, then the Taurus passes, with peaks above 3 kilometers.
- From early December to April the passes were closed by snow, and the trade stopped.



Risk on the road

- Safety came from treaties with the local kings, who undertook to police the roads.
- No payments to armed escorts appear in the caravan accounts.
- Robbery is rare in the texts, but never absent.
- Sacrifices before every trip, and a sheep at dangerous river crossings.

An incantation for the road: *"The black dog lurks on a mound, waiting for the scattered caravan, looking out for a good man."*

Arriving at Kanesh

- Straight to the palace on the mound: the goods are opened and cleared before anything goes down to the lower town.
- Tax on textiles was 3 parts in 60, which the Assyrians themselves sometimes called **5 percent**.
- Tax on tin: 2 pounds per bag, on a standard bag of 65 pounds.
- The palace could also buy up to 10 percent of the textiles at a reasonable price, before anyone else.
- The king used the fine textiles as gifts for his officials and allies.

Two ways to sell

- Cash sale, when the boss in Assur needs money now: *“until the silver has entered, you must not hand over the tin and textiles!”*
- Credit sale: an agent takes the goods and pays later, at about 50 percent above the Kanesh market price.
- Terms ran from a few weeks to seventy weeks; three months was common.
- Unpaid agent debts (*ba'abtum*) were a permanent drain, and the men in Kanesh spent their time chasing them.

Trust

- The instruction repeats: find an agent *“one who is as secure as you yourself!”*
- Agents delayed, lied, and disappeared, and everybody was traveling all the time.
- A merchant had most of his assets in other people's hands, and no double-entry bookkeeping.
- So he had **“only a fairly hazy idea of the actual state of his business and finances”**.
- The Arabic saying Larsen quotes: *“Love each other, but make accounts with each other.”*

One full turn

- **Stage 1, Kanesh.** The cargo is sold. Enlil-bani holds 30 pounds of silver, 15 kilos, and hands it sealed to Kukkulanum.
- **Stage 2, six weeks home.** *One bag on one donkey: silver travels light.*
- **Stage 3, Assur.** The silver becomes cargo: 114 textiles, 183 pounds of tin, 6 black donkeys, plus harness, tolls, and half a pound to each packer as working capital.
- **Stage 4, six weeks out.** The packers walk beside the donkeys, selling textiles of their own along the road. *The same trade at one sixtieth the scale, on the firm's silver.*
- **Stage 5, Kanesh again.** The palace takes its tax, the goods bring 45 to 50 pounds, the packers repay their half pound, and the silver waits for the next trip.

The Kukkulanum account, Larsen ch. 14. Goods move one way, silver the other.

One caravan's accounts

Purchases:	Silver	
114 <i>kutani</i> textiles	7½ pounds,	4¼ shekels
3 talents, 3 minas of tin	13⅝ pounds,	2⅝ shekels
6 donkeys	2 pounds,	8 shekels
<i>Investments</i>	23½ pounds,	5½ shekels
Working capitals	1 pound,	7 shekels
Other expenses:		
donkey harness		16 shekels
tin for expenses	2⅝ pounds,	2⅝ shekels
clothes for packers		4 shekels
miscellaneous		15 shekels
export toll		15 shekels
<i>Expenses</i>	4 pounds,	9⅙ shekels
Non-commercial expenses		
for Assur-malik		6 shekels
for Kukkulanum	⅝ pounds	
Extras	⅝ pounds,	6 shekels
Total spent in Assur	29⅓ pounds,	¼ shekel

Kukkulanum carries 30 pounds of silver from Kanesh to Assur; the men there spend it and send the cargo back. Larsen, ch. 14.

Reading the account

- Purchases (114 *kutānu* textiles, tin, 6 donkeys) take more than two-thirds of the silver.
- Harness, tin for the road, clothes for the packers and tolls take about an eighth, or about 16 percent counting the payments to individuals.
- Packers were paid in working capital (*be'ulatum*): money they traded on their own account and repaid at the end.
- Larsen's parallel is the parable of the talents. Wages here are the right to do business with someone else's money.

Why it paid

- Tin bought in Assur at about 13 shekels of tin per shekel of silver; sold in Kanesh at 7.
- An ordinary *kutānu* textile: under 4 shekels in Assur, 10 to 12 in Kanesh, 15 to 20 through a credit agent.
- A shipment costing about 30 pounds of silver in Assur was worth 45 to 50 in Kanesh.
- Profit above **50 percent**, before Anatolian taxes and expenses.
- Silver was cheaper in Anatolia than in Assur, and that gap is what the system ran on.

How much moved

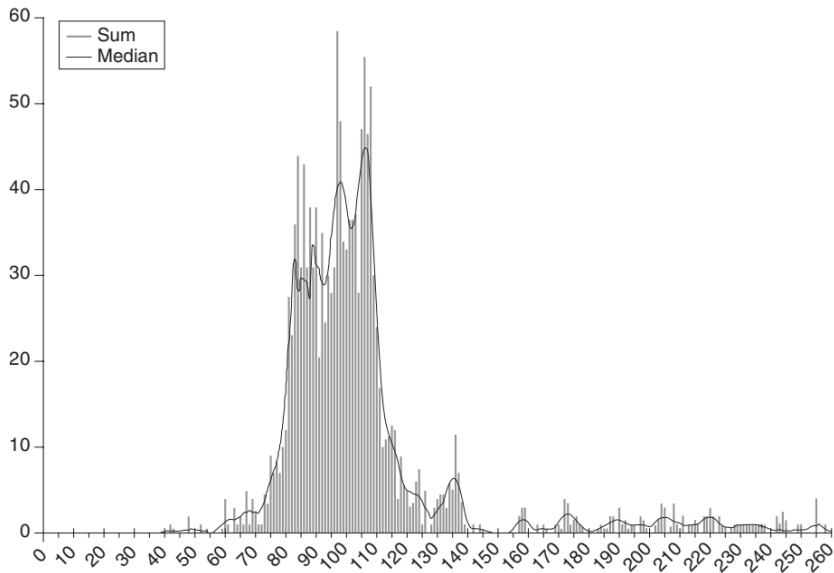
- Counting is hard: most texts are unpublished, and more are still in the ground.
- Veenhof in 1972 counted 14,500 textiles and 27,000 pounds of tin, from fewer than 200 texts.
- Scaled up to the excavated corpus: nearly **4 tons of tin** and about 3,800 textiles a year.
- That is roughly 110 donkey-loads of textiles and 55 of tin, every year.
- At Anatolian prices, more than a ton of silver a year.
- One family, Assur-idi's: perhaps three donkey-loads a year. A large system of small firms.

Where the goods came from

- Tin reached Assur from the south, from what the texts call the Lower Lands, probably through Susa.
- Its origin is most likely north-eastern Afghanistan, Uzbekistan and Tajikistan, possibly further east.
- That eastern leg is invisible: no texts, no archaeology, for a trade running to many tons.
- Fine textiles probably came mostly from southern Mesopotamia; some were woven in the households of Assur.
- One standard piece, 4 by 4.5 meters, is about 150 days of one woman's work.

The copper circuit inside Anatolia

- Copper was mined in the Pontic region and refined at Durhumit, which became the main copper market.
- It traveled west by ox-cart through Wahshushana to Purushaddum, and was exchanged there for silver and wool.
- Copper was hardly ever shipped back to Assur: too heavy for its value, and the Taurus was in the way.
- So the notion that the Assyrians came to Anatolia to get copper for Assur is, in Larsen's word, misleading.
- 4 tons of tin a year implies close to 40 tons of bronze made in Anatolia.



17. Graph of the dated texts plotted on a grid showing the number of texts and the sequence of eponymies. From Barjamovic, Hertel and Larsen 2012.

One generation of paperwork

- The dated texts pile into a narrow window: a sudden rise around eponym year 80, a crash around 110.
- More than 90 percent of the texts fall inside about thirty years.
- In calendar terms, roughly 1895 to 1865 BC.
- So what we can see in this detail is one generation of traders, not two centuries.
- Why the record collapses, decades before the fire, is the next class.

From letters to data

TRADE, MERCHANTS, AND THE LOST CITIES OF THE BRONZE AGE*

GOJKO BARJAMOVIC

THOMAS CHANEY

KEREM COŞAR

ALI HORTAÇSU

We analyze a large data set of commercial records produced by Assyrian merchants in the nineteenth century BCE. Using the information from these records, we estimate a structural gravity model of long-distance trade in the Bronze Age. We use our structural gravity model to locate lost ancient cities. In many cases, our estimates confirm the conjectures of historians who follow different methodologies. In some instances, our estimates confirm one conjecture against others. We also structurally estimate ancient city sizes and offer evidence in support of the hypothesis that large cities tend to emerge at the intersections of natural transport routes, as dictated by topography. Finally, we document persistent patterns in the distribution of city sizes across four millennia, find a distance elasticity of trade in the Bronze Age close to modern estimates, and show suggestive evidence that the distribution of ancient city sizes, inferred from trade data, is well approximated by Zipf's law. *JEL* Codes: F1, N7, N9, R12.



Turning tablets into trade data

- The paper starts from about 23,500 texts excavated at Kültepe, roughly 12,000 of them deciphered and edited.
- Transliterated into the Latin alphabet, so city names can be searched by machine.
- 2,806 tablets mention at least two cities.
- Those were then read one by one, to separate real shipments from cities named in passing.
- What survives the filter: 198 tablets.

What a data point looks like

- An expense memo: *“I paid 2 shekels of silver and 2 shekels of tin for the hire of a donkey from Timelkiya to Hurama.”*
- An itinerary: *“From Durhumit until Kanesh I incurred expenses of 5 minas of refined (copper), I spent 3 minas of copper until Wahsusana”.*
- A trip from A to B to C is counted as three journeys: A to B, B to C, A to C.
- 227 itineraries yield 391 directed journeys between pairs of cities.

These are exactly the caravan accounts Larsen reads in ch. 14, read for a different purpose.

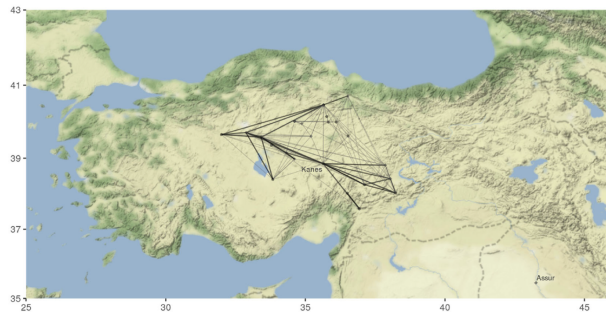
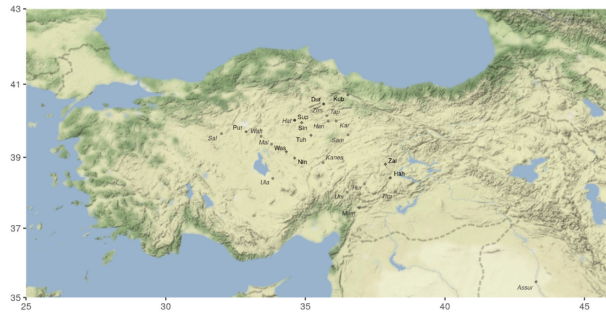
The sample

Object	Number
Known cities	15
Lost cities	10
Tablets with commercial itineraries	198
Itineraries identified	227
Journeys identified	391
Possible directed pairs	600
Pairs with any journey	114
Mean distance between known cities	223 km

- Small by modern standards, and unmatched for the second millennium BC.

What is counted

- The number of journeys, not the value of the cargo: the texts rarely say what was in the bags.
- So the measure is *“a count of the number of shipments”*.
- The matrix is sparse: 114 of the 600 possible directions have any journey at all.
- Where a pair does appear, the average is a little over three journeys.



Known cities and lost cities

- Of the 25 Anatolian cities in the sample, 15 are located on the ground and 10 are not.
- A **lost city** is named in the tablets, traded with and taxed, and has no site on the map.
- Historians have proposed sites for them from itineraries, surviving place-names and terrain.
- Kanesh sits in the middle of the network, but the flows are not simply a hub with spokes.

Reading distance off trade is not straightforward

- Large cities trade a lot with everyone, near and far.
- Small cities look distant because they trade little with anyone.
- A city ringed by many suppliers buys less from each one of them.
- Size and position have to be separated before the flows say anything about a map.

Gravity

Gravity in one line

$$Trade_{ij} = G \frac{Size_i \times Size_j}{Distance_{ij}^{\zeta}}$$

- Bigger places trade more.
- More distant places trade less.
- ζ says how fast trade falls off with distance.
- The same relation fits modern data on countries, regions and firms.

From textbook gravity to the paper

Textbook version

- trade in value;
- GDP or population observed;
- distances observed;

This paper

- counts of journeys;
- economic size not observed;
- some distances unknown;

The model has to estimate elasticity, size and coordinates at the same time.

From gravity to shares

Every flow into j obeys the same rule: $Trade_{kj} = G Size_k Size_j / Distance_{kj}^\zeta$.

$$s_{ij} = \frac{Trade_{ij}}{\sum_k Trade_{kj}} = \frac{G Size_j \cdot Size_i / Distance_{ij}^\zeta}{G Size_j \cdot \sum_k Size_k / Distance_{kj}^\zeta} = \frac{Size_i / Distance_{ij}^\zeta}{\sum_k Size_k / Distance_{kj}^\zeta}$$

- Sum every flow into j , then divide: G and $Size_j$ sit top and bottom, and cancel.
- What survives: j 's purchases split in proportion to each supplier's pull, size discounted by distance. A new supplier near j eats everyone else's slice.
- Behind the slice formula sits a full model: merchants buy wherever the delivered cost is lowest, and this is what that competition adds up to.

The paper derives the same slice formula from first principles; the derivation is in the backup slides.

Fitting the map to the flows

- Unknowns: one decay rate ζ , one size per city, and the coordinates of the 10 lost cities.
- The 15 known cities anchor the exercise: on their pairs, trade and distance are both visible, and that is where the decay rate is learned.
- Each lost city is then moved on the map until the trade its position predicts matches the trade the tablets record.
- A lost city that appears mostly with northern partners gets pulled north, unless being small, or far from everyone, explains its flows better.

A triangulation, like a GPS fix: known anchors, plus a known relation between signal and distance.

The estimated version

$$s_{ij}^{model} = \frac{Size_i Distance_{ij}^{-\zeta}}{\sum_{k \neq j} Size_k Distance_{kj}^{-\zeta}}, \quad s_{ij}^{data} = \frac{N_{ij}^{data}}{\sum_{k \neq j} N_{kj}^{data}}$$

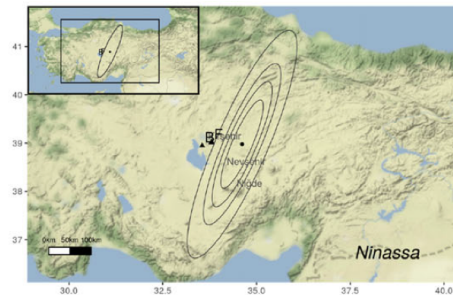
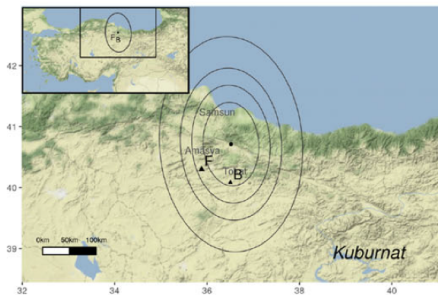
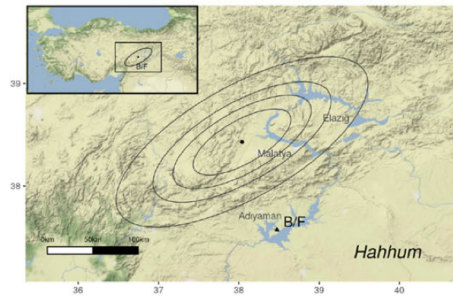
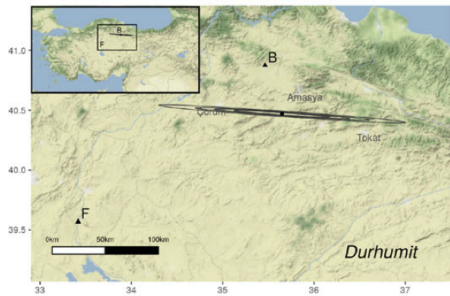
- Total trade is never observed, only the 391 recorded journeys. So the fit is on shares: of the journeys arriving in j , the fraction coming from i .
- On the left, the fraction the equation predicts; on the right, the fraction the tablets record.
- With the distance rule ζ learned on the known pairs, the computer chooses the sizes and the coordinates of the lost cities to minimize

$$\sum_j \sum_{i \neq j} \left(s_{ij}^{data} - s_{ij}^{model} \right)^2.$$

How fast trade fell with distance

- The estimate: $\hat{\zeta} = 1.912$. Trade falls roughly with the square of distance.
- Measured on the known pairs only, where distance is observed.
- Modern overland trade among Turkish cities gives a figure around 2.
- Four thousand years apart, and the same order of magnitude.

This is the earliest estimate of the gravity relation there is, about four millennia before the next one.



How to read the maps

- The filled dot is where the trade flows put the city.
- The rings around it show how tightly the flows pin it down.
- F and B are the proposals of the historians Forlanini and Barjamovic, made from the texts and the terrain.
- The small box in each corner places the panel on the map of Anatolia; modern towns are named in gray for orientation.
- The historians' proposals are not used in the estimation, so agreement between them is not built in.

Durhumit

- The copper market of the network, named more than 200 times in the Kanesh records, and lost.
- Its location is disputed: Assyrian and Hittite sources point in different directions.
- The flows put it **48 km** from Barjamovic's proposal, and 220 km from Forlanini's.
- Among unexcavated sites near the estimate, one carries an Assyrian seal cut in a workshop at Kanesh.

TABLE III
GRAVITY ESTIMATION RESULTS

Panel A: Distance elasticity and statistics

ζ (dist. elast.)	1.912
	(0.189)
Observations	600

Panel B: Sizes and locations of lost cities

	$Pop_i T_i^{\frac{1}{\theta}}$	(Std. err.)	Precision	Distance to historians' proposals, in km			
				Forlanini	(Std. err.)	Barjamovic	(Std. err.)
Durhumit	0.174	(0.409)	49	220	(46)	48	(26)
Hahhum	64.556	(85.929)	62	102	(27)	102	(27)
Kuburnat	11.22	(24.316)	76	72	(46)	70	(41)
Ninašša	0.21	(0.45)	87	71	(47)	93	(49)
Purušhaddum	0.076	(0.1)	154	168	(75)	193	(112)
Šinahuttum	1.515	(2.021)	34	24	(20)	24	(20)
Šuppiluliyā	0.012	(5.246)	89,914	89	(54,307)	85	(54,240)
Tuhpiya	0.579	(0.912)	38	128	(35)	112	(33)
Wašhaniya	5.413	(6.462)	35	68	(25)	13	(19)
Zalpa	28.695	(60.689)	145	103	(75)	131	(70)

City by city

- Washaniya: 13 km from Barjamovic's proposal.
- Sinahuttum: the two historians agree, and the estimate falls 24 km away.
- Ninassa: the flows favor Forlanini (71 km) over Barjamovic (93 km).
- Kuburnat: the two proposals lie about 70 km apart, and the estimate is about 70 km from each.
- Precision is better than 100 km for all but three cities, against a mean distance of 223 km between known cities.

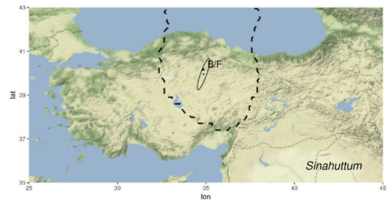
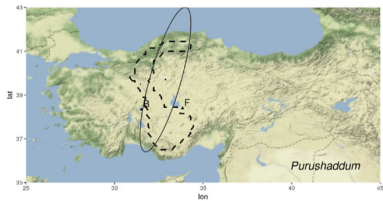
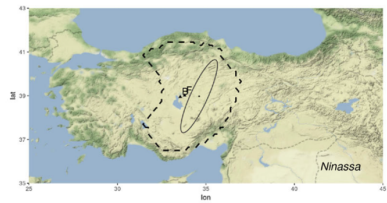
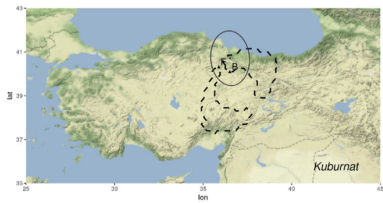
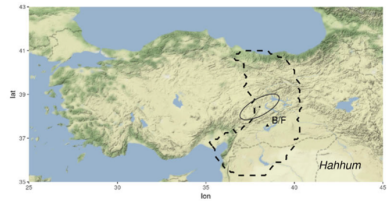
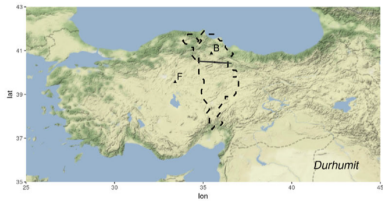
Where it does not work

- Hahhum comes out in the Taurus range: high, rugged, snow-covered, no place for a city.
- Nothing in the method knows that terrain is hostile. The historians impose it, and put Hahhum on the Euphrates.
- Tuhpiya lands far from both proposals, but 22 km from an older one by Cornelius in 1963.
- Purushaddum, Suppiluliyah and Zalpa are too imprecise to settle. The stated precision for Suppiluliyah is 89,914 km.



Losing a city on purpose

- A direct test: take a city whose site is known, hide the location, and recover it from the journeys alone.
- Each arrow runs from the city's true site to the spot the journeys recover: a short arrow means the method found it.
- Central cities come back close: Hattus 3 km off, Tapaggas 17 km, Hanaknak 19 km, Kanesh 33 km.
- Across the nine central cities the average error is 40 km.
- Cities at the edge of the network, such as Shalatuwar and Wahshushana, come back far less precisely.
- More neighbors means more angles to triangulate from.



Checking against the itineraries

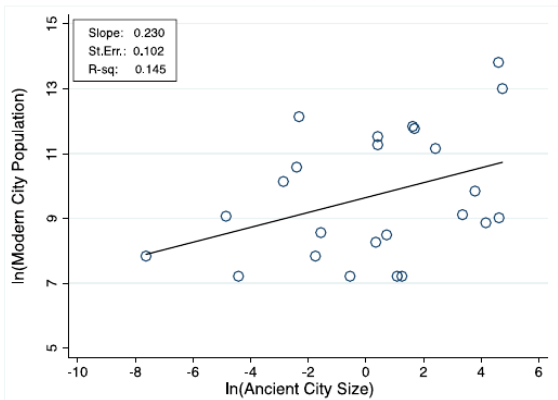
- Some texts give the order of stops, which restricts where a city in the middle of the route can be.
- Those constraints become admissible regions on the map, using no historian's coordinates.
- In each map, the dashed line is where the itineraries allow the city to sit; the solid ellipse is where the trade flows put it.
- Every estimate falls inside its region, with one exception.
- The exception is Purushaddum, at the western edge of the Assyrian zone, where the network gives least to work with.
- Where its ellipse and its dashed region overlap sits an unsurveyed mound, Akşehir Karahöyük: a guess a shovel could test.

City sizes

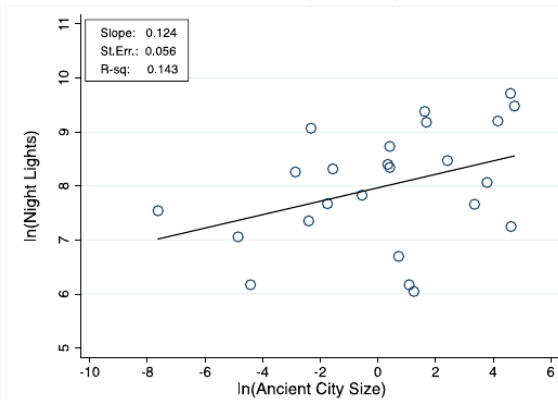
Size, without a census

- No population figures survive for any of these cities, and none can be dug up.
- The same fit that places the cities also says how much each one supplies, once distance is netted out.
- That becomes a measure of economic size, corrected for the advantage of sitting centrally.
- Kanesh, Mamma and Timelkiya come out at the top; Karahna and Zimishuna at the bottom.

Panel A: Population



Panel B: Night Lights



Four thousand years later

- Ancient size is compared with two modern measures: urban population within 20 km, and night lights.
- Both track the ancient estimate, with a correlation of 0.38.
- Ankara is dropped as an outlier: a minor provincial town until it was made capital in 1923.
- Large ancient cities tend to sit under large modern ones, which is also why some may never be excavated.

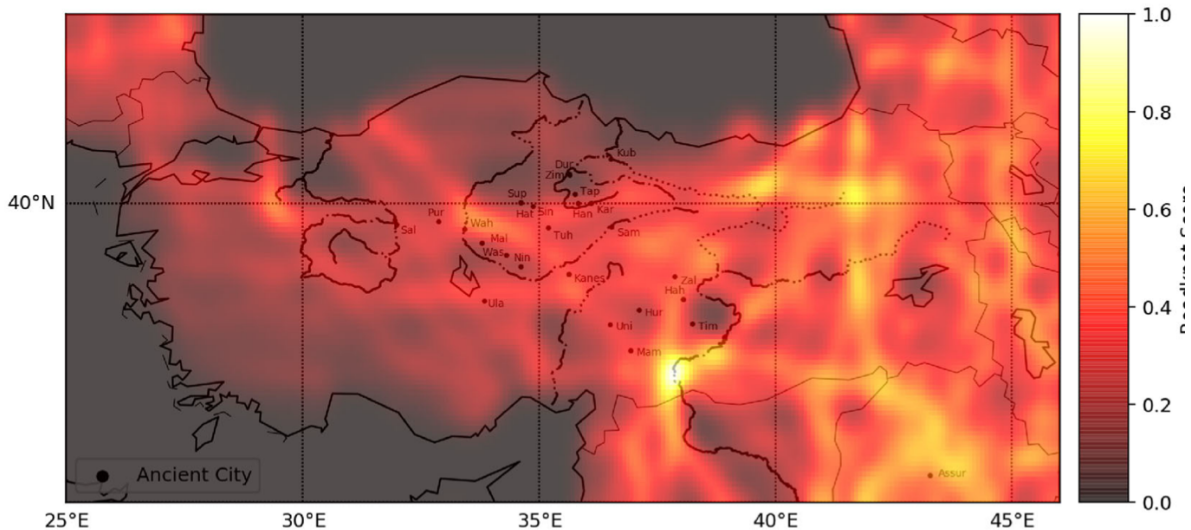
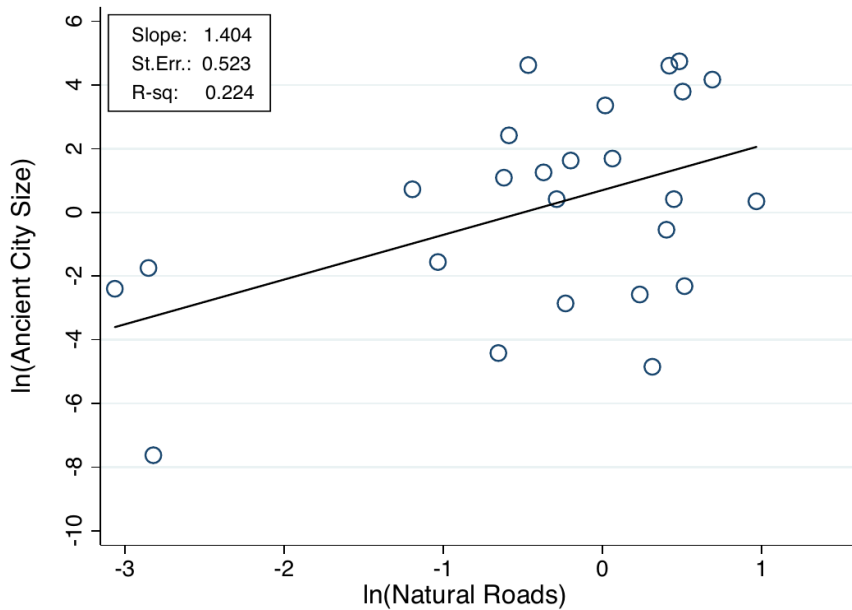


TABLE IV
DETERMINANTS OF ANCIENT CITY SIZES

	$\log \left(PopT^{\frac{1}{\theta}} \right)$				
	(1)	(2)	(3)	(4)	(5)
$\log (NaturalRoads)$	1.404** (.013)			1.783*** (.002)	
$\log (RomanRoads)$		1.990 (.378)			2.387 (.220)
$\log (Ruggedness)$			2.371** (.012)	3.189*** (.000)	2.495*** (.003)
N	25	25	25	25	25
R^2	0.224	0.038	0.166	0.508	0.220

Natural roads

- Ramsay in 1890: the terrain allows only a few ways across Anatolia, and cities grow where those ways cross.
- The paper rebuilds the ways from topography alone: walking times over rough ground, river fords, coastal sailing.
- Cheapest paths between very many origin and destination pairs, then a count of where paths cross or overlap.
- Bright places on the map are heavily crossed. No modern population or road data goes in.



Road knots and city size

- Cities near many crossings of natural routes are the large ones.
- That measure on its own accounts for about **22 percent** of the variation in ancient size.
- Rugged terrain nearby also goes with larger cities, which suggests a defensive advantage.
- Crop yield, rivers, minerals do not predict size.
- Roman roads point the same way but more weakly, and roads may follow cities rather than the reverse.

The knot outlasts the city

- Kanesh was abandoned; Kayseri, the regional capital today, is 20 km away.
- The same pattern holds for Hurama and Elbistan, Mamma and Maraş, Samuha and Sivas.
- Anatolia is a good place to test this, because the mountains leave few options. On a plain, topography says much less.
- Closer to home: the towns of the Brazilian interior are strung along the colonial cattle and gold roads and their river crossings.

Takeaways

- Kanesh is the earliest trade we can read from the traders' own paperwork.
- It ran on a price gap, on treaties that kept the roads open, and on trust in agents at a distance.
- Counted rather than read, the same records behave like modern trade data, and fall off with distance at close to the modern rate.
- Counting them puts lost cities back on the map, mostly where historians had already put them.
- At the edges of the network, where there is little to triangulate from, the method has little to say.

Backup: the model behind the equation

The environment

- There are $K + L$ cities: K known and L lost.
- Tradable goods are indexed by ω : tin, copper, wool, textiles and so on.
- Merchants arbitrage price differences between cities.
- There are iceberg costs: to deliver 1 unit from i to j , one has to ship $\tau_{ij} \geq 1$ units.
- There is also an internal cost $\tau_{jj} \geq 1$: supplying one's own city and its hinterland also costs something.
- Production and supply costs vary by city and good.

These are the goods Larsen describes: tin and textiles coming from Assur, copper and wool within Anatolia, silver as payment.

When does i sell to j ?

$$\tau_{ij}c_i(\omega) < \tau_{jj}c_j(\omega)$$

- The merchant buys where the delivered cost is low and sells where the local price is high.
- City j buys from whoever delivers at the lowest cost:

$$p_j(\omega) = \min_k \{\tau_{kj}c_k(\omega)\}.$$

- Who wins that market depends on the whole distribution of costs, not on an isolated pair.

Costs

$$\Pr[c_i(\omega) \leq c] = 1 - \exp\left(-T_i w_i^{-\theta} c^\theta\right)$$

- T_i : the efficiency or fundamental supply advantage of city i .
- w_i : the cost of local factors.
- θ : a parameter for the dispersion of costs.
- For each good, some city is relatively cheap; merchants exploit those differences.
- The parameters T_i , w_i and θ describe the city and hold for all goods; ω appears only in the cost draw $c_i(\omega)$.

Three steps to $E[N_{ij}]$

- **Delivered cost.** The production cost of i plus the freight to j define the *strength* of i in the market of j :

$$\phi_{ij} = T_i w_i^{-\theta} \tau_{ij}^{-\theta}.$$

High when i is efficient, has cheap factors and is close to j .

- **Who wins the market.** The chance that i supplies j is the strength of i over the sum of the strengths of all suppliers to j :

$$\pi_{ij} = \frac{\phi_{ij}}{\sum_k \phi_{kj}}.$$

- **From chance to count.** With many goods, that chance becomes the expected fraction of shipments arriving in j ; multiplying by the size of j , we obtain $E[N_{ij}]$.

The numerator on the next slide is exactly ϕ_{ij} , the strength of i : the cost parameter, after freight.

The structural gravity equation

$$E[N_{ij}] = N \frac{T_i w_i^{-\theta} \tau_{ij}^{-\theta}}{\sum_k T_k w_k^{-\theta} \tau_{kj}^{-\theta}} X_j$$

- The numerator rises when i is efficient and cheap to deliver to j .
- The denominator compares i with all the other possible suppliers of j .
- N : the number of goods, which scales the counts; X_j : total expenditure of j .
- In equilibrium, the trade balance gives $X_i = w_i L_i$: the income of i is paid to local factors.
- The form is structural: it comes from a least-cost decision.

Multilateral resistance

$$E[N_{ij}] = N \frac{\overbrace{T_i w_i^{-\theta} \tau_{ij}^{-\theta}}^{\text{strength of } i}}{\underbrace{\sum_k T_k w_k^{-\theta} \tau_{kj}^{-\theta}}_{\text{competition in } j}} X_j$$

- The denominator is **multilateral resistance**: how much i has to compete for the market of j with all the other suppliers.
- If j is close to many suppliers, that denominator is large and i sells less there, even with a small τ_{ij} .
- Bilateral distance matters, but always *relative* to the rest of the network.

Trade cost and distance

$$\tau_{ij}^{-\theta} = \delta \text{Distance}_{ij}^{-\zeta}$$

- The working assumption: bilateral frictions can be summarized by geographic distance.
- ζ is estimated from the data.
- If ζ is high, small differences in distance generate large differences in journeys.
- This relationship turns counts of journeys into information about the map.

δ is just a scale constant, the same for all pairs: it cancels when we use shares, so it affects neither ζ nor the locations.

The distance elasticity: ζ

$$\tau_{ij}^{-\theta} = \delta \text{Distance}_{ij}^{-\zeta} \quad \implies \quad E[N_{ij}] \propto \text{Distance}_{ij}^{-\zeta}$$

- ζ is the distance elasticity: it governs how fast trade falls with distance.
- Distance enters trade *only* through the exponent ζ , whatever θ may be. That is why it is ζ that the data identify ($\hat{\zeta} = 1.912$), measured on the pairs of known cities, where distance is observed.

θ (cost dispersion) is not identified by the data; the authors use $\theta = 4$ from the modern trade literature. It only shifts the scale of the sizes, not the results.